February 17, 2018



Little Tiger Invitational

Thermodynamics

Team Name: _____Team Number:_____

Score:_____/39

Olympian Names:_____

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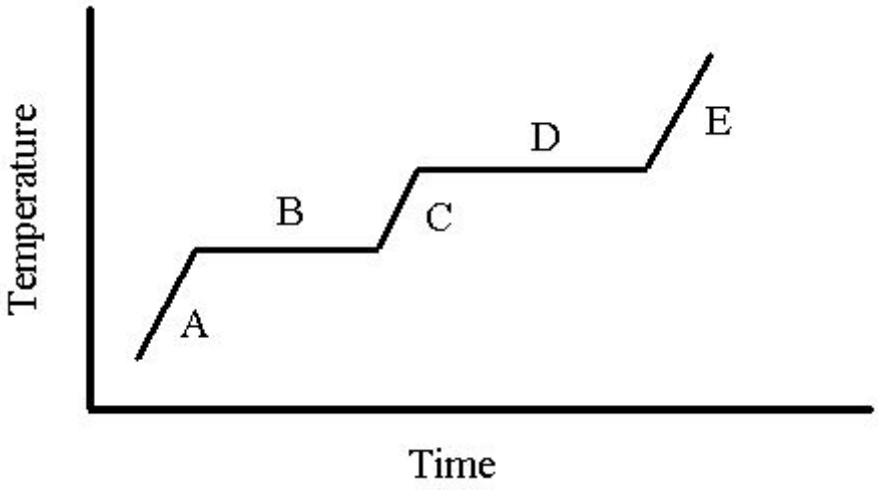
Multiple Choice (1 point each)

1. Which of the following is the most reasonable value for the temperature of this room in Kelvin?
    a. 68 K
    b. 293 K
    c. 20 K
    d. 200 K
2. Absolute zero is what on the Fahrenheit scale?
    a. -460 °F
    b. 0 °F
    c. 32 °F
    d. -32 °F
3. Imagine a tank of hydrogen gas initially at standard temperature and pressure (273 K, 1.00 atm) in a closed container. Which of the following necessarily occur when the temperature of the gas increases by 10.0 K?
    a. The magnitude of the total force that all of the gas molecules exert on the wall of its container decreases.
    b. The hydrogen changes phases, becoming a supercritical fluid.
    c. The velocity of each gas molecule increases.
    d. The average kinetic energy of the gas molecules increases.
4. 1 _____ is the quantity of heat required to raise the temperature of 1 g of liquid water by 1 K.
    a. calorie
    b. kilojoule
    c. joule
    d. BTU
5. All of the following are equivalent to 1 J EXCEPT
    a. $\frac{kg \cdot m^2}{s^2}$
    b. $N \cdot m$
    c. $10^{-3} kJ$
    d. $\frac{N \cdot kg}{s}$

6. Any thermodynamically versed American can tell you that normal human body temperature is 98.6 °F. An English chap, however, would likely tell you that body temperature is …
    a. 37 °C
    b. 273 K
    c. 32 °C
    d. 29 °C
7. Consider the following hypothetical situation: You are a rice fanatic and are frustrated at the current time that it takes for you to cook your rice (in boiling water) on the stove. To cook your rice faster, you could …
    a. Turn up the burner
    b. Move to the beach (at a lower altitude than your current home)
    c. Move to the mountains (at a higher altitude than your current home)
    d. Two of the above
8. How much heat is required to raise the temperature of 30.0 kg of liquid water from 15 °C to 95 °C (c=4186 J/kg · K) ?
    a. $1.2 \times 10^7\ J$
    b. $1.0 \times 10^7\ J$
    c. $8.0 \times 10^5\ J$
    d. $3.3 \times 10^6\ J$
9. How much heat is needed to melt 2.0 kg of ice initially at 0.°C? ($L_F = 333\ kJ/kg\ ,\ L_V = 2260\ kJ/kg$)
    a. $7 \times 10^5\ J$
    b. $5 \times 10^6\ J$
    c. $8 \times 10^3\ J$
    d. None of the above
10. Heat is
    a. A measure of the total energy of all the molecules in an object
    b. A measure proportional to the average kinetic energy of the molecules of a system
    c. A measure that refers to the transfer of energy from one object to another because of a difference in temperature
    d. Measured in Kelvin
11. The Second Law of Thermodynamics most directly deals with which concept?
    a. Entropy
    b. Enthalpy
    c. Heat
    d. Temperature

12. Natural processes tend to move toward a state of greater
    a. Order
    b. Disorder
    c. Temperature
    d. Happiness
13. A steel block and a wood block have been sitting in a room for a few hours. Which block feels colder to the touch? Why?
    a. The steel block. It's temperature is less than that of the wood block.
    b. The steel block. Heat transfers from your hand to the block at a faster rate than that of the wood block.
    c. The wood block. It's temperature is less than that of the steel block.
    d. The wood block. Heat transfers from your hand to the block at a faster rate than that of the steel block.
14. One end of a 33-cm-long aluminum rod $(k = 200\ J/s \cdot m \cdot K)$ with a diameter of 2.0 cm is kept at 460 °C, and the other is immersed in water at 22 °C. Calculate the heat conduction rate along the rod. (Hint: Be sure to use SI units in your calculation)
    a. 330 J/s
    b. 84 J/s
    c. 8.3×10^5 J/s
    d. 1.7×10^4 J/s
15. If A is in thermal equilibrium with B and B is in thermal equilibrium with C, then
    a. A is in thermal equilibrium with C
    b. A has a higher temperature than B
    c. B has a higher temperature than A
    d. None of the above
16. The phenomenon in question 15 is known as the
    a. 0th Law of Thermodynamics
    b. 1st Law of Thermodynamics
    c. 2nd Law of Thermodynamics
    d. 3rd Law of Thermodynamics
17. If a system absorbs 284.3 J of heat and does 446.9 J of work on its surroundings, the change in the internal energy of the system is
    a. 1.271×10^5 J
    b. - 731.2 J
    c. -162.6 J
    d. 162.6 J

18. The phenomenon in question 17 is known as the
    a. 0th Law of Thermodynamics
    b. 1st Law of Thermodynamics
    c. 2nd Law of Thermodynamics
    d. 3rd Law of Thermodynamics



Questions 19 - 22 refer to the above phase change diagram.

19. Which segment represents melting?
    a. A
    b. B
    c. C
    d. D
    e. E

20. Which segment represents boiling?
    a. A
    b. B
    c. C
    d. D
    e. E

21. Only the vapor phase is present on which segment?
    a. A
    b. B
    c. C
    d. D
    e. E

22. An appropriate unit for the vertical axis could be
    a. K
    b. J
    c. W
    d. m
23. An isobaric process is a thermodynamic process in which
    a. Volume is constant
    b. Temperature is constant
    c. Pressure is constant
    d. Number of moles is constant
24. Under an _____ process, no work can be performed.
    a. Isobaric
    b. Adiabatic
    c. Isothermal
    d. Isochoric
25. The most efficient possible heat engine is called the
    a. Carnot engine
    b. Joule engine
    c. Otto engine
    d. Internal combustion engine
26. _____ formulated the second law of thermodynamics and the entropy as equation of state.
    a. Clausius
    b. Joule
    c. Count Rumford
    d. Helmholtz
27. An English Brewer, James Prescott Joule an experiment involving _____ to determine that a given amount of work done was always equivalent to a particular amount of heat input.
    a. Falling weights and water
    b. Cannon boring
    c. Burning food
    d. Fire
28. The 0th Law of Thermodynamics was postulated _____ the 1st Law of Thermodynamics
    a. Before
    b. After
    c. Simultaneously as

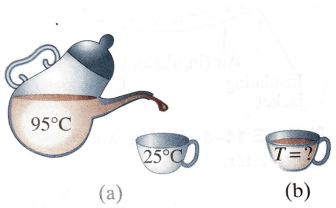
29. _____ defined an absolute temperature scale based on the Carnot cycle
    a. Sadi Carnot
    b. Anders Celsius
    c. Daniel Gabriel Fahrenheit
    d. Lord Kelvin
30. _____stated the second law of thermodynamics in the following form: "It is impossible to construct an engine which, working in a complete cycle, will produce no effect other than the raising of a weight and the cooling of a heat reservoir."
    a. Clausius
    b. Carnot
    c. Max Planck
    d. Lord Kelvin

CONTINUE TO FREE RESPONSE

Open Ended: (3 pts. each)

USE SIG FIGS AND METRIC UNITS! (the metric unit of temperature is °C)
Only answers written on the answer sheet will be scored.
Use scientific notation if necessary.

31. If 200 mL of tea (c=4186 J/kg · K) at 95 °C is poured into a 150 g glass cup (c=840 J/kg · K) initially at 25 °C, what will be the common final temperature of the tea and cup when equilibrium is reached, assuming no heat flows to the surroundings? (Hint: the heat lost by the tea is equal to the heat gained by the cup)



32. How much heat is needed to melt 16.50 kg of silver that is initially at 20. °C? ($c = 230\ J/kg \cdot K$, $L_F = 88\ kJ/kg$, $L_V = 2300\ kJ/kg$)

33. A concrete highway ($\alpha = 12 \times 10^{-6} \ (^{\circ}C)^{-1}$) is built of slabs 12 m long. How long should the expansion slabs be (at 20 °C) to prevent buckling if the range of temperatures is -30 °C to 50 °C?